

CSEDU Digital Knowledge Platform

Comprehensive User Manual

Department of Computer Science and Engineering
University of Dhaka

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1 Introduction

1.1 Overview

The CSEDU Digital Knowledge Platform is a comprehensive web-based system designed to facilitate knowledge sharing, research collaboration, and library management within the Computer Science and Engineering Department at the University of Dhaka. The platform integrates multiple services including a digital library, research repository, student project showcase, multimedia archives, and AI-powered assistance.

1.2 Key Features

- **Digital Library Management** - Browse, borrow, and return books with automated fine calculations
- **Research Repository** - Submit, review, and publish academic research papers
- **Project Showcase** - Display and archive student projects with team collaborations
- **Digital Archives** - Store and access historical documents, multimedia content
- **AI Assistant** - Intelligent search and document summarization using RAG technology
- **Role-Based Access Control** - Five-tier permission system ensuring appropriate access
- **Notification System** - Email and in-app notifications for important events

1.3 User Roles

The platform supports five distinct user roles, each with specific permissions:

1. **Public (Unauthenticated)** - View published content only
2. **Student** - Default role with basic borrowing and upload permissions
3. **Researcher** - Enhanced access to submit and review research papers
4. **Librarian** - Manage library catalog, approve content, process loans
5. **Administrator** - Full platform management and user role administration

1.4 Accessing the Platform

Platform Access

URL: <https://devops.farefin.com/>

2 Getting Started

2.1 Creating an Account

2.1.1 Registration Process

1. Navigate to the platform homepage
2. Click on the **"Sign Up"** or **"Register"** button
3. Fill in the registration form:
 - **Full Name:** Your complete name as per university records
 - **Email Address:** A valid email address (preferably institutional email)
 - **Password:** Minimum 8 characters, include uppercase, lowercase, numbers, and special characters
 - **Confirm Password:** Re-enter your password
4. Click **"Create Account"**
5. Your account is created with **Public** role by default

Important Note

All new registrations are assigned the **Public** role automatically. To request elevated permissions (Researcher or Librarian), use the Role Request feature after logging in.

2.2 Logging In

1. Click the **"Login"** button on the homepage
2. Enter your registered email address
3. Enter your password
4. Click **"Sign In"**
5. You can also login using magic link which will go to your email address.
6. You will be redirected to your personalized dashboard

2.3 Profile Management

2.3.1 Viewing Your Profile

1. Click on your name/avatar in the top-right corner
2. Select **"Profile"** from the dropdown menu
3. View your profile information:

- Name, Email, Role
- Registration date
- Last login time
- Account statistics (uploads, loans, etc.)

3 Library Catalog System

3.1 Browsing the Catalog

The library catalog is accessible to all users, including non-authenticated visitors.

3.1.1 Accessing the Catalog

1. Click "**Library**" or "**Catalog**" in the main navigation menu
2. The catalog page displays all available books in a grid or list view
3. Each book card shows:
 - Book title and author
 - Cover image (if available)
 - Availability status (Available/Borrowed)
 - Format (Book, E-book, etc.)
 - Year of publication

3.1.2 Searching for Books

1. Use the search bar at the top of the catalog page
2. Enter keywords from:
 - Book title
 - Author name
 - ISBN number
 - Topic/subject area
3. Press Enter or click the search icon
4. Results are displayed with relevance ranking

3.1.3 Filtering Books

You can narrow down catalog results using multiple filters:

- **Topic:** Filter by subject area (e.g., Algorithms, Machine Learning, Databases, Programming)
- **Availability:** Show only available or borrowed books
- **Format:** Book, E-book, Reference
- **Year:** Publication year range

3.2 Borrowing Books (Student, Researcher, Librarian, Administrator)

3.2.1 How to Borrow

1. Log in to your account
2. Navigate to the library catalog
3. Find the book you want to borrow (must show "Available")
4. Click on the book to view details
5. Click the "**Borrow**" button
6. Confirm the borrowing action
7. A success message appears with the due date
8. The book is now listed in "**My Loans**"

3.2.2 Loan Period and Limits

- **Loan Duration:** 20 hours from the checkout date
- **Due Date:** Displayed in your loans list
- **Late Returns:** Subject to fines (50 BDT per day, max 500 BDT per loan)
- **Fine Block:** If outstanding fines exceed 50 BDT, borrowing is disabled until payment

Borrowing Restrictions

- You cannot borrow the same book twice simultaneously
- Librarians cannot borrow books (monitoring role only)
- Borrowing is disabled if you have overdue fines exceeding 50 BDT

3.3 Viewing Your Loans

1. Click "**My Loans**" in the navigation menu
2. See all your current and past loans
3. For each loan, view:
 - Book title
 - Checkout date
 - Due date
 - Return date (if returned)
 - Status (Active, Returned, Overdue)
4. Active loans show a "**Return**" button

3.4 Returning Books

1. Go to "**My Loans**"
2. Find the book you want to return
3. Click the "**Return Book**" button
4. Confirm the return
5. If the book is overdue, a fine will be automatically calculated
6. The book status changes to "Returned"
7. Book availability is updated in the catalog

3.5 Reservations (Holds) System

3.5.1 Placing a Hold

When all copies of a book are borrowed, you can place a hold to be notified when it becomes available.

1. Navigate to the book detail page
2. If all copies are borrowed, a **"Place Hold"** button appears
3. Click **"Place Hold"**
4. Your hold request is recorded with a timestamp
5. You'll receive an email notification when the book becomes available
6. The hold is valid for 48 hours after notification

3.5.2 Viewing Your Holds

1. Click **"My Loans"** or **"Holds"** in the navigation
2. View all your active reservations
3. See your position in the queue
4. Status indicators:
 - **Active:** Waiting for book availability
 - **Fulfilled:** Book is now available for you
 - **Cancelled:** You or staff cancelled the hold

3.5.3 Cancelling a Hold

1. Go to your holds list
2. Find the hold you want to cancel
3. Click the **"Cancel Hold"** button
4. Confirm cancellation
5. The hold is removed from your list

3.6 Fine Management

3.6.1 Viewing Your Fines (Student, Researcher)

1. Click **"Fines"** in the navigation menu
2. See all your fines with details:
 - Book title

- Fine amount (BDT)
- Status (Pending, Paid, Waived)
- Calculation date

3. Total unpaid fines are displayed at the top

3.6.2 Fine Calculation

- **Rate:** 50 BDT per day overdue
- **Maximum:** 500 BDT per loan
- **Calculation:** Automatic on the day after due date
- **Grace Period:** None - fines start immediately after the due date

3.6.3 Paying Fines

1. Navigate to "**Fines**" page
2. Find the pending fine you want to pay
3. Click "**Pay Fine**"
4. Select payment method (if configured)
5. Confirm payment
6. Fine status changes to "Paid"
7. If total fines were blocking borrowing, the block is lifted

Payment Note

Contact the library desk or administrator for payment procedures. The system records payment confirmations entered by library staff.

4 Digital Archives

4.1 Overview

The Digital Archives section stores historical documents, multimedia content, and departmental resources accessible to all users.

4.2 Browsing Archives

1. Click **"Archives"** in the main navigation
2. Browse through categories or search
3. Each archive item displays:
 - Title
 - Description/Abstract
 - Upload date
 - Format (PDF, Video, Image, Document, etc.)
 - Access tier (Public, Student, Researcher, Restricted)
4. Click on an item to view full details

4.3 Uploading to Archives (Student, Researcher, Librarian, Administrator)

4.3.1 Supported File Types

- **Documents:** PDF, DOCX, DOC, PPTX, PPT, XLSX, XLS
- **Images:** JPG, JPEG, PNG, GIF
- **Videos:** MP4
- **Audio:** MP3
- **Archives:** ZIP
- **External Links:** URLs (YouTube videos, web resources)
- **Maximum File Size:** 50 MB per file

4.3.2 Upload Process

1. Click **"Upload"** in the navigation
2. Select **"Archive"** as the upload type
3. Fill in the form:
 - **Title:** Descriptive title (required)

- **Abstract:** Brief description of the content
 - **Keywords:** Comma-separated tags for searchability
 - **Tags:** Additional categorization tags
 - **Language:** Content language (default: English)
 - **Access Tier:** Choose who can view (Public, Student, Researcher, Restricted)
 - **Status:** Draft (save for later) or Published (make available immediately)
4. Choose file source:
 - Upload a file from your computer, OR
 - Provide an external URL (for YouTube videos, web resources)
 5. Click **"Upload"**
 6. Wait for processing confirmation
 7. Your upload appears in **"My Uploads"**

4.4 Viewing Archive Details

1. Click on any archive item
2. View complete details:
 - Title, abstract, keywords
 - Uploader name and date
 - File format and size
 - Access permissions
3. Options available:
 - **Download:** Save file to your device
 - **View Online:** Preview (for PDFs, images, videos)
 - **Share:** Copy link to share with others

4.5 Downloading Archive Files

1. Navigate to the archive item detail page
2. Click the **"Download"** button
3. The file downloads to your default download folder
4. For external links, you're redirected to the external resource

4.6 Managing Your Uploads

1. Click **"My Uploads"** in the navigation
2. View all your uploaded content (Archives, Research, Projects)
3. For each item:
 - View status (Draft, Review, Published, Archived)
 - Edit metadata (title, abstract, keywords)
 - Delete drafts
 - Submit for review/publication
4. Click **"Edit"** to update metadata
5. Click **"Delete"** to remove draft items (published items require admin approval)

5 Research Paper Management

5.1 Overview (Researcher, Librarian, Administrator Only)

The Research section allows researchers to submit, review, and publish academic papers, facilitating knowledge sharing and collaboration.

5.2 Browsing Research Papers (All Users)

1. Click **"Research"** in the navigation menu
2. Browse published research papers
3. Use search and filters:
 - Search by title, author, keywords
 - Filter by publication year
 - Filter by research area/topic
4. Each paper displays:
 - Title and authors
 - Publication date
 - Abstract preview
 - DOI (if available)
 - Download option

5.3 Submitting Research Papers (Researcher, Administrator)

5.3.1 Preparing Your Submission

Before submitting, ensure you have:

- Final version of your paper (PDF format, max 50 MB)
- Complete list of authors with affiliations
- Abstract (required)
- Keywords (5-10 recommended)
- DOI, journal/conference details (if published)

5.3.2 Submission Process

1. Click **"Upload"** then select **"Research Paper"**
2. Fill in the submission form:
 - **Title:** Full paper title (required)
 - **Authors:** Enter all authors' names (comma-separated)

- **Co-authors:** Additional contributors
 - **Abstract:** Concise summary of your research (required)
 - **Keywords:** Key terms for discoverability
 - **DOI:** Digital Object Identifier (if available)
 - **Journal/Conference:** Publication venue
 - **Publication Date:** When published (if applicable)
 - **Access Tier:** Choose visibility (Public, Student, Researcher)
3. Upload your PDF file
 4. Choose initial status:
 - **Draft:** Save for later editing
 - **Submit for Review:** Send to reviewers immediately
 5. Click "**Submit Research Paper**"
 6. Confirmation message appears
 7. Your submission is listed in "**My Uploads**"

5.4 Paper Review Workflow

5.4.1 Submission Statuses

1. **Draft:** Author is still editing
2. **Under Review:** Assigned to reviewer for evaluation
3. **Published:** Approved and publicly available
4. **Archived:** Withdrawn or removed from active display

5.4.2 Submitting for Review

1. Go to "**My Uploads**"
2. Find your draft paper
3. Click "**Submit for Review**"
4. Confirm submission
5. Status changes to "Under Review"
6. Librarian/Administrator assigns a reviewer
7. You'll be notified when review is complete

5.4.3 Receiving Feedback

1. Check your notifications or email
2. Navigate to your paper in **"My Uploads"**
3. Click to view review comments
4. Review notes include:
 - Reviewer feedback
 - Suggestions for improvement
 - Publication decision
5. If revisions requested:
 - Make necessary changes
 - Re-upload revised version
 - Resubmit for review

5.5 Viewing Published Papers

1. Click on any published paper
2. View full details:
 - Complete abstract
 - Author information
 - Publication details (journal, DOI, date)
 - Full-text PDF
3. Download the PDF
4. Cite the paper (citation format provided)
5. Use AI to generate summary

5.6 Editing Your Papers

1. Go to **"My Uploads"**
2. Find the paper you want to edit
3. Click **"Edit"**
4. Update metadata (title, abstract, keywords, etc.)
5. Replace PDF file if needed
6. Save changes
7. If already published, changes may require admin approval

6 Student Projects

6.1 Overview (Student, Researcher, Administrator)

The Projects section showcases student team projects, providing a portfolio of academic and personal work.

6.2 Browsing Projects (All Users)

1. Click **"Projects"** in the navigation
2. Browse the project gallery
3. Filter by:
 - Academic year
 - Course code
 - Project type

4. Each project card shows:

- Project title
- Team members
- Supervisor
- Academic year
- Brief description

6.3 Submitting a Project (Student, Researcher, Administrator)

6.3.1 Before Submission

Prepare the following:

- Project title
- Team member names (all contributors)
- Supervisor name (optional)
- Course code and academic year
- Project abstract/description
- Keywords
- Project deliverables:
 - Documentation (PDF, DOCX)
 - Source code (ZIP archive)
 - Demo video (MP4 or YouTube link)
 - Web URL (if applicable)
 - GitHub repository link
 - APK file (for Android apps)

6.3.2 Submission Steps

1. Click "**Upload**" then select "**Project**"

2. Fill in project details:

- **Title:** Project name (required)
- **Team Members:** All member names (comma-separated, required)
- **Supervisor:** Faculty supervisor (select from list)
- **Academic Year:** Year of completion (e.g., 2024, required)
- **Course Code:** Related course (e.g., CSE400)
- **Abstract:** Project summary (required)
- **Keywords:** Tags for searchability

3. Add project resources:

- Upload project file (PDF, ZIP, APK)
- Web URL (for live demo)
- GitHub repository link
- App download link

4. Choose status:

- **Draft:** Save for later
- **Submit for Approval:** Send to librarian/admin

5. Click **"Submit Project"**

6. Your project appears in **"My Uploads"** with status **"Review"**

6.4 Project Approval Process

1. After submission, librarian/administrator reviews your project
2. Review criteria:
 - Completeness of information
 - Appropriate content
 - Working links and files
3. Project status changes:
 - **Approved** → **Published**: Visible in gallery
 - **Rejected** → **Draft**: Returned with feedback
4. You receive email notification of the decision
5. If rejected, make requested changes and resubmit

6.5 Viewing Project Details

1. Click on any project in the gallery
2. View complete information:
 - Full description and abstract
 - Team member list
 - Supervisor information
 - Academic year and course
 - Keywords and tags
3. Access project resources:
 - Download documentation
 - Visit web demo
 - View source code on GitHub
 - Download APK (for apps)

7 AI-Powered Features

7.1 AI Chat Assistant (Student, Researcher, Librarian, Administrator)

7.1.1 Overview

The AI Chat Assistant uses Retrieval-Augmented Generation (RAG) technology to answer questions by searching through all indexed documents in the platform (research papers, archives, projects).

7.1.2 Accessing the AI Chat

1. Click the **AI Chat icon** (usually in bottom-right corner)
2. Or navigate to **"AI Assistant"** in the menu
3. The chat interface opens with a welcome message

7.1.3 Asking Questions

1. Type your question in the chat input box
2. Questions can be in:
 - English
 - Bengali
 - Other supported languages (auto-detected)
3. Example questions:
 - "What research papers discuss machine learning algorithms?"
 - "Summarize projects related to Android development"
 - "Find documents about computer networking"
 - " ?" (Bengali)
4. Press Enter or click the Send button
5. Wait for the AI to process (usually 2-5 seconds)

7.1.4 Understanding AI Responses

AI responses include:

- **Answer:** Generated response based on indexed documents
- **Source Citations:** Links to documents used to generate the answer
- **Model Used:** Which AI model processed your query (Groq/Gemini)
- **Detected Language:** The language of your question
- **Response Time:** How long it took to generate

AI Capabilities

- Searches across all published content you have access to
- Respects access control (won't show restricted content)
- Provides cited sources for transparency
- Remembers context within a conversation session
- Can rephrase questions for better results

7.1.5 Viewing Source Documents

1. Click on any citation link in the AI response
2. You're taken to the source document details page
3. Read the full document or download it
4. Return to chat to continue conversation

7.2 Document Summarization

7.2.1 Generating Summaries

1. Navigate to any research paper or document
2. Click the **"Summarize"** button
3. Select summary options:
 - Language (English, Bengali, etc.)
 - Model (Groq or Gemini)
 - Length (Short, Medium, Long)
4. Click **"Generate Summary"**
5. Wait for processing (10-30 seconds depending on document length)
6. Read the AI-generated summary
7. Copy or share the summary

Summary Features

- Extracts key points from long documents
- Maintains technical accuracy
- Available in multiple languages
- Saves time on literature review

8 Role Request System

8.1 Requesting a Role Upgrade (Student, Researcher)

8.1.1 When to Request

Users can request role upgrades when they need additional permissions:

- **Student → Researcher:** To submit and review research papers
- **Student/Researcher → Librarian:** To manage library catalog and approve content

Note: Administrator role cannot be self-requested and must be assigned by existing administrators.

8.1.2 Submitting a Request

1. Click on your profile icon
2. Select "**Request Role Upgrade**"
3. Or navigate to **"/role-request"**
4. Fill in the request form:
 - **Desired Role:** Select Researcher or Librarian
 - **Justification:** Explain why you need this role (required)
 - Example justifications:
 - "I am a PhD student in the CSE department (ID: 2021-XX-XXX) and need to submit research papers"
 - "I am a faculty member and want to help manage the library catalog"
 - "I am working as a research assistant under Prof. X"
5. Click "**Submit Request**"
6. All administrators receive email notifications
7. Your request status appears on the page

8.1.3 Request Status

- **Pending:** Waiting for administrator review (you cannot submit another request while one is pending)
- **Approved:** Your role has been upgraded - log out and back in to see new permissions
- **Declined:** Request was not approved - see administrator notes for reason

8.1.4 Viewing Your Request History

1. Go to the role request page
2. Scroll down to see **"Your Requests"** section
3. View all past requests with:
 - Requested role
 - Submission date
 - Status (Pending, Approved, Declined)
 - Your justification
 - Administrator decision notes (if available)

9 Notifications

9.1 In-App Notifications

9.1.1 Accessing Notifications

1. Look for the **bell icon** in the top navigation bar
2. Red badge shows number of unread notifications
3. Click the bell to open notification dropdown
4. See recent notifications with:
 - Notification title
 - Brief message
 - Timestamp
 - Read/unread status

9.1.2 Notification Types

You receive notifications for:

- **Role Requests:**
 - Your request was approved/declined
 - (Admins) New role request submitted
- **Library:**
 - Book hold is now available for pickup
 - Loan due date approaching
 - Overdue book reminder
- **Research:**

- Paper review completed
 - Paper published
- **Projects:**
 - Project approved/rejected
- **System:**
 - Platform updates
 - Maintenance announcements

9.1.3 Managing Notifications

1. Click on any notification to:
 - Mark it as read
 - Navigate to related page
2. Click **”Mark all as read”** to clear all unread notifications
3. Click **”View All”** to see complete notification history

9.2 Email Notifications

All in-app notifications are also sent to your registered email address. Configure email preferences in Settings.

10 Librarian Features

10.1 Overview (Librarian, Administrator Only)

Librarians have additional permissions to manage the library catalog, process loans, handle fines, and approve content submissions.

10.2 Managing Library Catalog

10.2.1 Adding Individual Books

1. Navigate to **"Library"** → **"Catalog"**
2. Click **"Add Book"** button
3. Fill in book details:
 - **Title:** Full book title (required)
 - **Author:** Author name(s) (required)
 - **ISBN:** ISBN-10 or ISBN-13 (optional but recommended)
 - **Topic:** Subject area (auto-detected from title or manually selected)
 - **Format:** Book, E-book, Reference
 - **Year:** Publication year
 - **Publisher:** Publishing company
 - **Total Copies:** Number of copies available
 - **Location:** Physical shelf location
 - **Cover Image:** Upload book cover (optional)
4. Click **"Add Book"**
5. Book is immediately added to the catalog
6. Available copies = Total copies initially

10.2.2 Bulk Import from CSV

1. Navigate to **"Admin"** → **"Catalog Management"**
2. Click **"Import CSV"**
3. Download the CSV template
4. Fill in the template with book data
5. CSV columns:
 - title, author, isbn, topic, format
 - year, publisher, total.copies, location
6. Upload your completed CSV file

7. Click **”Import”**

8. System processes the file:

- Existing books (matched by ISBN) are updated
- New books are created

9. View import summary (successes, errors, duplicates)

10.2.3 Exporting Catalog

1. Navigate to **”Admin”** → **”Catalog Management”**

2. Click **”Export Catalog”**

3. Select export format (CSV)

4. Choose filters (optional):

- By topic
- By availability
- By format

5. Click **”Export”**

6. CSV file downloads to your device

7. Use for backup, analysis, or sharing

10.3 Managing Loans and Returns

10.3.1 Viewing All Loans

1. Navigate to **”Library”** → **”All Loans”**

2. See complete loan list with:

- Borrower name and email
- Book title
- Checkout date
- Due date
- Status (Active, Returned, Overdue)

3. Filter by status:

- Active loans
- Overdue loans
- Returned loans

4. Sort by due date, checkout date, or borrower name

10.3.2 Processing Returns

1. In the All Loans list, find the loan to process
2. Click "**Process Return**"
3. Confirm the return
4. System automatically:
 - Marks loan as returned
 - Calculates fines if overdue
 - Updates book availability
 - Notifies next user in hold queue (if any)

10.4 Fine Management

10.4.1 Viewing All Fines

1. Navigate to "**Library**" → "**All Fines**"
2. View complete fine list showing:
 - Borrower name and email
 - Book title
 - Fine amount (BDT)
 - Status (Pending, Paid, Waived)
 - Calculation date
3. See total unpaid fines at the top
4. Filter by status
5. Search by borrower name

10.4.2 Recording Fine Payments

1. Find the fine in the All Fines list
2. Click "**Record Payment**"
3. Confirm payment details:
 - Amount paid
 - Payment method
 - Reference number (if applicable)
4. Click "**Confirm**"
5. Fine status changes to "Paid"
6. Borrower is notified

10.4.3 Waiving Fines

1. Find the fine to waive
2. Click "**Waive Fine**"
3. Enter waiver reason (required):
 - "First-time offence"
 - "System error"
 - "Special circumstances"
 - Other explanation

4. Click **"Waive"**
5. Fine status changes to "Waived"
6. Action is logged in audit trail
7. Borrower is notified

10.5 Content Approval

10.5.1 Reviewing Submitted Content

1. Navigate to **"Admin"** → **"Content Review Queue"**
2. See all items pending approval:
 - Research papers awaiting review
 - Student projects for approval
 - Archive items for publishing
3. Click on any item to view details
4. Review content for:
 - Appropriate content
 - Complete metadata
 - Proper formatting
 - Working files/links

10.5.2 Approving/Rejecting Content

1. After reviewing, choose action:
2. **To Approve:**
 - Click **"Approve"** or **"Publish"**
 - Add approval notes (optional)
 - Confirm
 - Content becomes publicly visible
 - Uploader is notified
3. **To Reject:**
 - Click **"Reject"** or **"Request Changes"**
 - Enter reason for rejection (required)
 - Suggest improvements
 - Confirm
 - Content returns to draft status
 - Uploader is notified with your feedback

11 Administrator Features

11.1 Overview (Administrator Only)

Administrators have full platform control, including user management, role assignment, and system configuration.

11.2 User Management

11.2.1 Viewing All Users

1. Navigate to **"Admin"** → **"Users"**
2. See complete user list with pagination
3. For each user, view:
 - Name and email
 - Current role
 - Registration date
 - Last login time
4. Search users by name or email
5. Filter by role

11.2.2 Changing User Roles

1. Find the user in the user list
2. Click on their row to expand details
3. Click **"Change Role"**
4. Select new role:
 - Student
 - Researcher
 - Librarian
 - Administrator
5. Add reason for change (recorded in audit log)
6. Click **"Confirm"**
7. User is notified of role change
8. Change is logged in audit trail

Role Change Restrictions

- You cannot change your own role (prevents accidental self-demotion)
- Administrator role should be granted carefully
- Role downgrades may remove user's access to content they uploaded

11.3 Role Request Management

11.3.1 Viewing Pending Requests

1. Navigate to "**Admin**" → "**Role Requests**"
2. See queue of pending requests
3. For each request, view:
 - Applicant name and email
 - Current role
 - Requested role
 - Justification provided
 - Submission date
4. Sort by date (oldest first recommended)

11.3.2 Approving Requests

1. Click on a request to view full details
2. Review the justification
3. Verify applicant credentials (if needed)
4. Click "**Approve**"
5. Add approval notes (optional)
6. Confirm approval
7. System immediately:
 - Updates user's role
 - Sends email notification to user
 - Creates audit log entry
 - Marks request as "Approved"
8. User must log out and back in to see new permissions

11.3.3 Declining Requests

1. Click on the request
2. Click **"Decline"**
3. Enter reason for decline (required):
 - "Insufficient justification"
 - "Role not appropriate for user"
 - "Additional information needed"
 - Other specific reason
4. Click **"Confirm"**
5. User receives email with your reason
6. Request is marked as "Declined"
7. User can submit a new request with updated justification

11.4 Audit Log

11.4.1 Viewing the Audit Trail

1. Navigate to "**Admin**" → "**Audit Log**"
2. See chronological list of all administrative actions
3. Each entry shows:
 - Actor (admin who performed action)
 - Action type (e.g., "role_change", "fine_waived")
 - Resource type (user, book, fine, etc.)
 - Resource ID
 - Timestamp
 - Additional details
4. Filter by:
 - Actor (specific admin)
 - Action type
 - Date range
 - Resource type
5. Search by resource ID

11.4.2 Audit Log Actions Tracked

- User role changes
- Fine waivers
- Role request approvals/declines
- Content status changes (publish, archive)
- Catalog imports
- User account modifications
- System configuration changes

Audit Log Purpose

The audit log provides:

- Accountability for administrative actions
- Security and compliance monitoring
- Troubleshooting support
- Historical record of platform changes
- Immutable, append-only log

11.5 System Configuration

11.5.1 Platform Settings

Administrators can configure:

- **Library Settings:**

- Default loan period (currently 14 days)
- Fine rate (currently 50 BDT/day)
- Maximum fine per loan (currently 500 BDT)
- Fine block threshold (currently 50 BDT)

- **Upload Settings:**

- Maximum file size (currently 50 MB)
- Allowed file types
- Default access tiers

- **AI Settings:**

- Preferred AI model
- Query limits
- Language preferences

- **Email Notifications:**

- SMTP configuration
- Email templates
- Notification preferences

Note: System configuration changes require environment variable updates and may require platform restart.

12 Search and Discovery

12.1 Global Search

1. Use the search bar in the top navigation
2. Enter keywords
3. Search across:
 - Library catalog (titles, authors, ISBN)
 - Research papers (titles, abstracts, authors)
 - Student projects (titles, descriptions, team members)
 - Digital archives (titles, content, keywords)
4. Results are grouped by type
5. Filter results by:
 - Content type
 - Date range
 - Access tier

12.2 Advanced Search

1. Click **"Advanced Search"**
2. Set multiple criteria:
 - Exact phrase matching
 - Author/creator name
 - Date range
 - Specific fields (title, abstract, keywords)
 - Boolean operators (AND, OR, NOT)
3. Select content types to search
4. Click **"Search"**
5. Refine results with faceted filters

12.3 Browsing by Category

1. Navigate to specific sections:
 - **Library:** Browse by topic, format, or year
 - **Research:** Browse by research area or year
 - **Projects:** Browse by academic year or course

- **Archives:** Browse by collection or media type
2. Use hierarchical filters to narrow results
 3. View item counts for each category

13 Troubleshooting and FAQs

13.1 Common Issues and Solutions

13.1.1 Login Problems

- **Forgot password:**

- Click "Forgot Password?" on login page
- Check email for reset link (check spam folder)
- Reset link expires in 1 hour

- **Account locked after failed attempts:**

- After 5 failed login attempts, account is locked for 15 minutes
- Wait for the lockout period to expire
- Or use password reset to unlock immediately

- **Email not registered:**

- Verify you're using the correct email
- Create a new account if needed

13.1.2 Upload Issues

- **File too large:**

- Maximum file size is 50 MB
- Compress PDF files or videos before uploading
- Consider using external links for large videos

- **Unsupported file type:**

- Check supported formats list
- Convert file to supported format

- **Upload timeout:**

- Check your internet connection
- Try uploading during off-peak hours
- Upload smaller files

13.1.3 Borrowing Issues

- **Cannot borrow (button disabled):**

- Check if you have outstanding fines exceeding 50 BDT
- Verify all copies aren't already borrowed
- Ensure you're logged in
- Librarians cannot borrow books

- **Book not available:**

- Place a hold to be notified when available
- Check estimated return date

13.1.4 Search Not Working

- **No results found:**

- Check spelling of search terms
- Try broader keywords
- Remove filters
- Use advanced search for more control

- **Incomplete results:**

- Ensure you have permission to view restricted content
- Check if content is published (not draft)

13.1.5 AI Chat Issues

- **AI not responding:**

- Wait up to 30 seconds for response
- Check your internet connection
- Try rephrasing your question
- Start a new chat session

- **Irrelevant answers:**

- Be more specific in your question
- Include context or keywords
- Try asking in a different language

13.2 Frequently Asked Questions

13.2.1 General

1. **Q: Can I use the platform without an account?**

A: Yes, you can browse published content (library catalog, research papers, projects) without logging in. However, borrowing books, uploading content, and using AI features require an account.

2. **Q: How do I get elevated permissions (Researcher/Librarian)?**

A: Use the Role Request feature from your profile. Provide a clear justification. An administrator will review and approve/decline your request.

3. **Q: Can I delete my uploaded content?**

A: You can delete draft items. Published content requires administrator approval for deletion to maintain platform integrity.

4. **Q: How long does content review take?**

A: Review times vary depending on administrator availability. Typically 1-5 business days. You'll receive email notification when review is complete.

13.2.2 Library

1. **Q: How many books can I borrow at once?**

A: There is no hard limit on simultaneous loans, but you must not have overdue fines exceeding 50 BDT.

2. **Q: Can I renew a loan?**

A: Currently, the system does not support automatic renewals. Return the book and re-borrow if available, or contact a librarian for assistance.

3. **Q: What if I lose a borrowed book?**

A: Contact the library immediately. Replacement cost will be assessed by the librarian.

4. **Q: Can fines be waived?**

A: Librarians and administrators can waive fines on a case-by-case basis with proper justification.

13.2.3 Research and Projects

1. **Q: Who can submit research papers?**

A: Only users with Researcher, Librarian, or Administrator roles can submit research papers.

2. **Q: Can I edit my research paper after submission?**

A: You can edit papers in draft status. After submission for review or publication, contact an administrator for changes.

3. **Q: Can multiple students collaborate on a project?**

A: Yes, list all team members in the project submission. The submitter becomes the primary contact.

4. **Q: How do I add a supervisor to my project?**

A: Select the supervisor from the user list during project submission. The supervisor must already have an account.

13.2.4 AI Features

1. **Q: What content does the AI search?**

A: The AI searches all published content you have permission to access, including research papers, projects, archives, and library catalog descriptions.

2. **Q: Can I trust AI responses?**

A: AI responses are based on indexed documents. Always verify critical information by checking the cited sources. Use AI as a starting point, not the final authority.

3. **Q: What languages does the AI support?**

A: The AI supports English and Bengali, with auto-detection. Additional languages may be available depending on configuration.

4. **Q: Are my chat conversations private?**

A: Chat sessions are associated with your account. Administrators do not routinely view chat logs, but conversations are stored for system improvement and support purposes.

14 Best Practices

14.1 For All Users

- **Use strong passwords:** Include uppercase, lowercase, numbers, and special characters
- **Keep your profile updated:** Ensure your email is current for notifications
- **Provide descriptive titles:** Use clear, searchable titles for uploads
- **Add keywords:** Enhance discoverability with relevant tags
- **Respect copyright:** Only upload content you have permission to share
- **Use appropriate access tiers:** Don't restrict public content unnecessarily
- **Cite sources:** When uploading research, properly attribute references

14.2 For Students

- **Return books on time:** Avoid fines and help others access materials
- **Place holds early:** Don't wait until you urgently need a book
- **Save drafts:** Work on projects incrementally before final submission
- **List all team members:** Ensure proper credit for collaborative work
- **Provide detailed abstracts:** Help others understand your project
- **Use meaningful project titles:** Avoid vague names like "Final Project"

14.3 For Researchers

- **Complete metadata:** Provide full author lists, keywords, and abstracts
- **Include DOI:** Add DOI when available for easier citation
- **Peer review thoughtfully:** Provide constructive feedback
- **Keep papers updated:** Submit revisions when significant changes occur
- **Collaborate openly:** Share preliminary findings when appropriate

14.4 For Librarians

- **Maintain catalog accuracy:** Double-check ISBN and publication details
- **Process returns promptly:** Keep availability information current
- **Document fine waivers:** Provide clear reasons in audit trail
- **Review content fairly:** Apply consistent approval standards
- **Respond to holds quickly:** Notify users when reserved books are available

14.5 For Administrators

- **Review role requests promptly:** Reduce waiting times for users
- **Document role changes:** Always include justification in audit log
- **Monitor system health:** Regularly check logs for errors
- **Backup data regularly:** Export catalogs and content lists periodically
- **Communicate changes:** Inform users of platform updates

15 Glossary

Access Tier

Permission level required to view content (Public, Student, Researcher, Librarian, Restricted)

Audit Log

Immutable record of all administrative actions on the platform

Citation Reference to a source document used by the AI to generate a response

DOI Digital Object Identifier - unique identifier for academic papers

Fine Monetary penalty for overdue book returns (50 BDT/day, max 500 BDT)

Hold Reservation on a borrowed book to be notified when it becomes available

ISBN International Standard Book Number - unique identifier for books

Loan Period

Duration for which a book can be borrowed (14 days)

Metadata

Descriptive information about content (title, abstract, keywords, tags)

RAG Retrieval-Augmented Generation - AI technology that searches documents to answer questions

Role Tier

User permission level (Public, Student, Researcher, Librarian, Administrator)

Session Continuous conversation thread in AI chat

Status Content publication state (Draft, Review, Published, Archived)

16 Appendices

16.1 Appendix A: Quick Reference Card

Quick Actions by Role

Student

- Browse catalog → Borrow book → Return → Pay fines
- Upload archive/project → Edit metadata
- Request role upgrade
- Use AI chat assistant

Researcher

- Everything students can do PLUS:
- Submit research papers
- Review other researchers' papers
- Access researcher-tier content

Librarian

- Everything researchers can do (except borrow) PLUS:
- Add books to catalog
- Import/export catalog
- View/manage all loans and fines
- Waive fines
- Approve content (projects, research, archives)

Administrator

- Everything librarians can do PLUS:
- Manage users and change roles
- Approve/decline role requests
- View audit logs
- Configure system settings

16.2 Appendix B: Keyboard Shortcuts

Action	Shortcut
Open search	Ctrl + K (Cmd + K on Mac)
Focus search bar	/
Open AI chat	Ctrl + . (Cmd + . on Mac)
Open notifications	Ctrl + N (Cmd + N on Mac)
Navigate to Library	Alt + L
Navigate to Research	Alt + R
Navigate to Projects	Alt + P
Navigate to Archives	Alt + A
Open profile	Alt + U
Logout	Alt + X

16.3 Appendix C: File Format Support

Category	Formats	Max Size
Documents	PDF, DOCX, DOC, PPTX, PPT, XLSX, XLS	50 MB
Images	JPG, JPEG, PNG, GIF	50 MB
Videos	MP4	50 MB
Audio	MP3	50 MB
Archives	ZIP	50 MB
Mobile Apps	APK	50 MB
External Links	URL (YouTube, websites)	N/A

16.4 Appendix D: API Endpoints (For Developers)

For Technical Users

This section is for developers integrating with the platform or building custom tools.

16.4.1 Authentication Endpoints

POST /api/v1/auth/register	- Create account
POST /api/v1/auth/login	- Login
POST /api/v1/auth/refresh	- Refresh access token
GET /api/v1/auth/me	- Get current user info
POST /api/v1/auth/logout	- Logout
POST /api/v1/auth/forgot-password	- Request password reset
POST /api/v1/auth/reset-password	- Reset password

16.4.2 Library Endpoints

GET /api/v1/library/catalog	- List catalog
GET /api/v1/library/catalog/{itemId}	- Get book details
POST /api/v1/library/loans	- Borrow book
GET /api/v1/library/loans	- List user's loans
POST /api/v1/library/loans/{id}/return	- Return book
GET /api/v1/library/fines	- List user's fines
POST /api/v1/library/fines/{id}/pay	- Pay fine
POST /api/v1/library/holds	- Place hold
GET /api/v1/library/holds	- List holds
DELETE /api/v1/library/holds/{id}	- Cancel hold

16.4.3 Media Endpoints

POST /api/v1/media/upload	- Upload file
GET /api/v1/media	- List media
GET /api/v1/media/{itemId}	- Get media details
GET /api/v1/media/{itemId}/download	- Download file
GET /api/v1/media/my-uploads	- List user's uploads
PATCH /api/v1/media/{itemId}/metadata	- Edit metadata
DELETE /api/v1/media/{itemId}	- Delete media

16.4.4 Research Endpoints

POST /api/v1/research	- Submit paper
GET /api/v1/research	- List papers
GET /api/v1/research/{paperId}	- Get paper details
PUT /api/v1/research/{paperId}	- Update paper
POST /api/v1/research/{id}/submit	- Submit for review
POST /api/v1/research/{id}/review	- Add review
POST /api/v1/research/{id}/publish	- Publish paper

16.4.5 AI Endpoints

POST /api/v1/ai/chat - Send chat query
GET /api/v1/ai/chat/history/{sessionId} - Get chat history
POST /api/v1/ai/summarize - Summarize document

16.4.6 Admin Endpoints

GET /api/v1/admin/users - List users
PATCH /api/v1/admin/users/{id}/role - Change user role
GET /api/v1/admin/audit-log - View audit log
GET /api/v1/admin/catalog/export - Export catalog CSV
POST /api/v1/admin/catalog/import - Import catalog CSV
GET /api/v1/admin/role-requests - List role requests
POST /api/v1/admin/role-requests/{id}/decide - Approve/decline

16.5 Appendix E: System Architecture Overview

16.5.1 Technology Stack

- **Frontend:** Next.js 14 (React framework), TypeScript, Tailwind CSS
- **Backend API:** Go (Golang), Chi router
- **Database:** PostgreSQL 16 with pgvector extension
- **File Storage:** MinIO (S3-compatible object storage)
- **Cache/Queue:** Redis
- **AI/RAG Service:** Python FastAPI, Sentence Transformers
- **LLM Providers:** Groq (primary), Google Gemini (fallback)
- **Web Server:** Nginx (reverse proxy)

16.5.2 Key Features

- JWT-based authentication with refresh token rotation
- Role-based access control (RBAC) at database and application levels
- Vector embeddings for semantic search (768-dimensional)
- Hybrid search combining full-text and semantic similarity
- Real-time notifications via WebSocket (planned)
- Audit logging for compliance and security
- Automated fine calculation via background worker
- Email notifications via SMTP (Brevo/Sendinblue)

16.6 Appendix F: Default User Accounts

For testing and initial setup, the following accounts are pre-configured:

Security Warning

Change these default passwords immediately in production!

Email	Password	Role
admin@cs.du.ac.bd	Admin@12345	Administrator
librarian@cs.du.ac.bd	Staff@12345	Librarian
researcher@cs.du.ac.bd	Research@12345	Researcher
student@cs.du.ac.bd	Student@12345	Student

16.7 Appendix G: Configuration Settings

16.7.1 Library Configuration

- **LOAN_PERIOD_DAYS:** 14 days
- **FINE_RATE_BDT_PER_DAY:** 50 BDT
- **MAX_FINE_PER_LOAN_BDT:** 500 BDT
- **FINE_BLOCK_THRESHOLD_BDT:** 50 BDT
- **FINE_CALC_INTERVAL:** 24 hours

16.7.2 Upload Configuration

- **Maximum File Size:** 50 MB per file
- **Allowed Document Formats:** PDF, DOCX, DOC, PPTX, PPT, XLSX, XLS
- **Allowed Media Formats:** MP4, MP3, JPG, JPEG, PNG, GIF
- **Allowed Archive Formats:** ZIP, APK

16.7.3 AI/RAG Configuration

- **Embedding Model:** sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2
- **Embedding Dimension:** 768
- **Chunk Size:** 512 tokens
- **Chunk Overlap:** 50 tokens
- **Top-K Results:** 10
- **Vector Search Limit:** 8
- **FTS Search Limit:** 8
- **Max Query Length:** 5000 characters
- **Max Context Window:** 10 previous messages

16.7.4 Authentication Configuration

- **JWT_EXPIRY_HOURS:** 1 hour
- **REFRESH_EXPIRY_DAYS:** 7 days
- **Bcrypt Cost:** 12 rounds
- **Failed Login Attempts:** 5 attempts
- **Account Lockout Duration:** 15 minutes
- **Password Reset Link Expiry:** 1 hour

16.8 Appendix H: Change Log

16.8.1 Version 1.0.0 (Initial Release)

Release Date: July 2026

Core Features:

- User authentication and authorization system
- Five-tier role hierarchy (Public, Student, Researcher, Librarian, Administrator)
- Library catalog management with search and filters
- Book borrowing and return workflow
- Automated fine calculation (50 BDT/day, max 500 BDT)
- Hold/reservation system
- Digital archive management
- Research paper submission and review workflow
- Student project showcase
- Media upload (PDF, DOCX, images, videos, etc.)
- AI-powered chat assistant with RAG
- Document summarization
- Role request system
- In-app and email notifications
- Audit logging for administrative actions
- CSV import/export for library catalog
- Mobile-responsive design

Security Features:

- JWT-based authentication
- Refresh token rotation
- Bcrypt password hashing (cost 12)
- Brute-force protection (5 attempts, 15-min logout)
- Role-based access control
- Audit trail for admin actions

Known Limitations:

- No automatic loan renewals

- Maximum file size limited to 50 MB
- No offline mode
- AI chat limited to English and Bengali
- No real-time WebSocket notifications (email only)

16.9 Appendix I: Roadmap (Planned Features)

16.9.1 Upcoming Features

- **Enhanced Search:**
 - Advanced Boolean operators
 - Saved search queries
 - Search history
- **Social Features:**
 - User profiles with bios
 - Follow other researchers
 - Comment on research papers
 - Rate and review books
- **Collaboration:**
 - Shared project workspaces
 - Real-time document editing
 - Discussion forums
- **Analytics:**
 - Usage statistics dashboard
 - Popular content reports
 - User engagement metrics
- **Mobile App:**
 - Native iOS and Android apps
 - Offline reading
 - Push notifications
 - Barcode scanning for checkouts
- **Integration:**
 - OAuth/SSO with university systems
 - Integration with institutional repositories
 - API for third-party applications

17 Conclusion

The CSEDU Digital Knowledge Platform represents a comprehensive solution for managing library resources, research papers, student projects, and digital archives within an academic environment. By integrating traditional library management with modern AI-powered features, the platform facilitates knowledge sharing, collaboration, and discovery.

17.1 Key Takeaways

- **Unified Access:** Single platform for all departmental knowledge resources
- **Role-Based Permissions:** Appropriate access levels for different user types
- **Intelligent Search:** AI-powered discovery across all content types
- **Streamlined Workflows:** Automated processes for borrowing, fines, and approvals
- **Transparency:** Audit logs and clear status tracking
- **Accessibility:** Mobile-responsive design, multilingual support

17.2 Getting the Most from the Platform

To maximize your experience:

1. Explore all sections to understand available resources
2. Use descriptive titles and keywords when uploading content
3. Leverage the AI assistant for quick information retrieval
4. Keep your profile updated for accurate notifications
5. Provide feedback to help improve the platform
6. Respect due dates to avoid fines and help others
7. Collaborate with colleagues through shared content

17.3 Continuous Improvement

The platform is actively maintained and regularly updated with:

- New features based on user feedback
- Security patches and bug fixes
- Performance optimizations
- Enhanced AI capabilities
- Expanded content collections

17.4 Your Feedback Matters

We encourage all users to:

- Report bugs and technical issues
- Suggest feature enhancements
- Share best practices with the community
- Contribute to documentation improvements
- Participate in user surveys

Thank you for using the CSEDU Digital Knowledge Platform!

For support, contact the CSE Department IT team

Department of Computer Science and Engineering CSEDU Digital Knowledge Platform
 User Manual 1 Introduction 1.1 Overview The CSEDU Digital Knowledge Platform is a comprehensive web-based system designed to facilitate knowledge sharing, research collaboration, and library management within the Computer Science and Engineering Department at the University of Dhaka. The platform integrates multiple services including a digital library, research repository, student project showcase, multimedia archives, and AI-powered assistance. 1.2 Key Features • Digital Library Management - Browse, borrow, and return books with automated fine calculations • Research Repository - Submit, review, and publish academic research papers • Project Showcase - Display and archive student projects with team collaborations • Digital Archives - Store and access historical documents, multimedia content • AI Assistant - Intelligent search and document summarization using RAG technology • Role-Based Access Control - Five-tier permission system ensuring appropriate access • Notification System - Email and in-app notifications for important events 1.3 User Roles The platform supports five distinct user roles, each with specific permissions: 1. Public (Unauthenticated) - View published content only 2. Student - Default role with basic borrowing and upload permissions 3. Dhaka-1000, Bangladesh